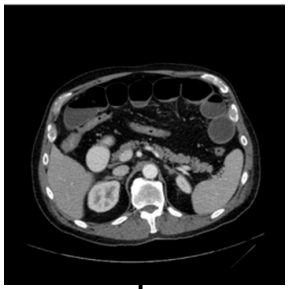


CT slice (224×224)



CNN encoder (ResNet-50)

$1/2 \cdot (64, 112, 112)$

$1/4 \cdot (256, 56, 56)$

$1/8 \cdot (512, 28, 28)$

$1/16 \cdot F (1024, 14, 14)$

CBAM

Channel Attention

Spatial Attention

+

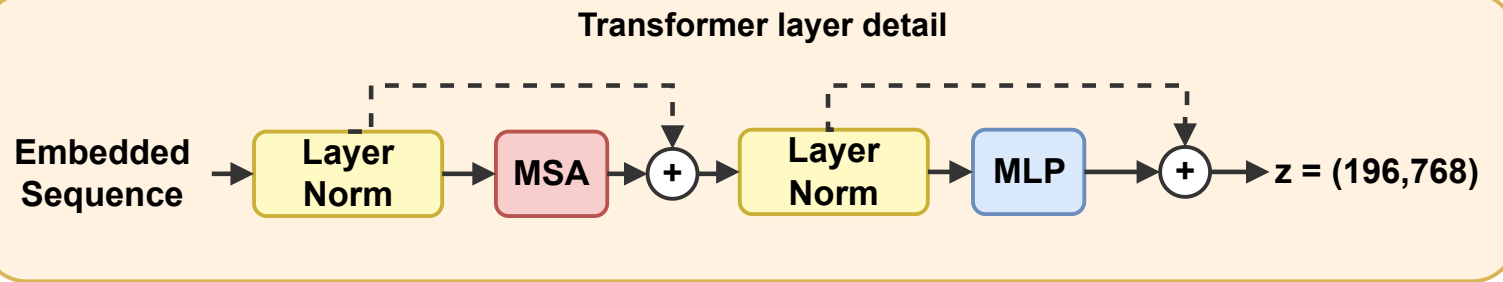
Refined hidden feature
(1024,14,14)

Linear Projection (Patch Embedding)

Transformer encoder

Transformer Layer $\times (n = 12)$

Hidden Feature
(196,768)



Conv 3×3 (16, 224, 224)

Segmentation head
Conv 3×3

Segmentation output

Up $\times 2$ (64, 224, 224)

Conv 3×3 (64, 112, 112)

Concat 3 @ $1/2 \rightarrow 192 \times 112 \times 112$

Up $\times 2$ (128,112,112)

skip 1/2
(64,112,112)

Conv 3×3 (128, 56, 56)

Concat 2 @ $1/4 \rightarrow 512 \times 56 \times 56$

Up $\times 2$ (256,56,56)

skip 1/4 (256,56,56)

Conv 3×3 (256, 28, 28)

Concat 1 @ $1/8 \rightarrow 1024 \times 28 \times 28$

Up $\times 2$ (512,28,28)

skip 1/8 (512,28,28)

Conv 3×3
(512,14,14)

Reverse Attention
(skip 1/8)

Reverse Attention module

Conv 3×3 (depthwise) + BN

reverse map = $1 - \sigma(\cdot)$

$\odot$ input feature

Bottleneck $1 \times 1 \rightarrow 3 \times 3 \rightarrow 1 \times 1$

$\oplus$ input feature $\rightarrow$ output

→ Conv 3×3 + ReLU

→ Upsample $\times 2$ (bilinear)

→ Segmentation head

--- Downsample (CNN encoder)

— Skip connection (CNN $\rightarrow$ decoder)

- - - Feature concatenation

⊕

Element-wise add — residual

Reverse Attention module